

Technical Report: Provided-Text Conditioned Audio Delivery Ranking for EmpathyEval Track 1

WhereIsAI-Lingnan

Final Release, July 2026

Abstract

This report documents the final Phase 1 submission system, `provided_text_conditioned_audio_delivery_ranking`. The system uses a single `AudioDeliveryPairwiseMLP` trained on all official labeled training data. Candidate decisions use prosody, `Emotion2Vec`, `WavLM`, task, and text-length metadata only. ASR, teacher models, model fusion, fallback rules, speech-quality features, and test-time training are excluded. The frozen submission covers 530 questions and has been reproduced byte-for-byte from the released final checkpoint and retained final feature inputs.

1 Public Artifacts and Integrity

The public code repository is <https://github.com/Beichen-Hg/WhereIsAI-Lingnan>. The public Hugging Face model repository is <https://huggingface.co/BertramM/whereisai-lingnan-track1-audio>. The Hugging Face repository contains only the final model release files: `model.pt`, `config.yaml`, `feature_group_importance.json`, the model card, and a SHA-256 manifest. Competition data, cached feature files, pretrained-model copies, and per-question test predictions are not publicly released in those repositories.

Artifact	SHA-256
Final checkpoint <code>model.pt</code>	24303cb353b767c6f8c964876beec 85cbf868633aa88e3f291cf8004a5e23083
Checkpoint configuration <code>config.yaml</code>	d26c614e9eb27207bd6c9acdc00419e 64e998ec32899410e159564af1d9d5038
Frozen submission JSONL	f84f1a12568f2556db273fc3b70e7d87 b6319e83cf086c7772405e3abd185222

Table 1: Final release integrity identifiers.

2 Final Method

2.1 Model

The final learned model is `AudioDeliveryPairwiseMLP`. For each question, the scorer constructs candidate-level representations and converts them into pairwise features for ranking the available response audios. The checkpoint contains the model state dictionary, the 1,775 feature names, and the fitted feature normalization statistics needed by inference.

The exact feature groups are:

- **Prosody**: basic acoustic delivery statistics extracted by the `librosa_basic` backend.
- **Emotion2Vec**: audio emotion embeddings from `emotion2vec/emotion2vec_plus_large`.
- **WavLM**: speech embeddings from `microsoft/wavlm-base-plus`.
- **Task**: task-type indicators.
- **Text metadata**: lengths of official JSON-provided context, user, and response text.

Candidate semantic text is taken from the official JSON field `json_provided`. No ASR transcript is used to override provided text. FunASR is only an optional loading backend for the Emotion2Vec emotion model; it is not used as an automatic-speech-recognition decision module.

2.2 Excluded Components

The final inference path rejects or does not invoke ASR semantic replacement, teacher calls, test labels, fusion models, fallback models, speech-quality feature groups, or test-time optimization. Dependency and submission audits are run as part of final inference.

2.3 Design Rationale and Decision Flow

The task is treated as a ranking problem rather than a response-generation problem. Given a shared conversational situation, the system selects the audio candidate whose delivery is most appropriate. Official text provides stable conditioning and length metadata only; it is not rewritten, semantically replaced by ASR, or scored by an LLM judge.

For each question, the system constructs user/candidate prosody, Emotion2Vec, WavLM, task, and text-metadata features. It then creates every unordered pair of candidates. The MLP estimates the probability that the left candidate is preferred to the right candidate from left, right, signed-difference, absolute-difference, and safe-ratio features. Each candidate accumulates its pairwise win probabilities, the score is normalized by the number of opponents, and the highest mean score is returned. This one mechanism handles both two-candidate and three-candidate questions without a separate model or a fusion rule.

Emotion2Vec and WavLM have complementary roles. Emotion2Vec contributes emotion-oriented audio representations, while WavLM contributes general speech representations. Prosody provides interpretable delivery statistics. The final MLP learns their user-to-candidate relationships together with the official task indicators and provided-text length metadata.

3 Training Protocol and Parameters

All official labeled training examples were used for the final refit. Phase 1 test examples were used only for blind manifest construction, feature extraction, inference, and submission auditing. No Phase 1 test labels are loaded by the final inference entry point.

The all-labeled training metrics are retained as training diagnostics and are not claimed as held-out generalization results.

Setting	Final value
Model	AudioDeliveryPairwiseMLP
Feature count	1,775
Hidden dimensions	[256, 128]
Dropout	0.15
Learning rate	0.0003
Weight decay	0.001
Batch size	128
Epochs	6
Random seed	2026
Training data	All official labeled training data

Table 2: Final refit configuration.

4 Runtime Environment

The final release was verified on Linux with Python 3.10.20, PyTorch 2.9.0+cu128, CUDA 12.8, NumPy 2.2.6, PyYAML 6.0.3, librosa 0.11.0, SoundFile 0.13.1, Transformers 5.8.1, Hugging Face Hub 1.14.0, FunASR 1.3.1, and ModelScope 1.36.3. GPU execution accelerates feature extraction; the final scorer also supports `DEVICE=cpu`. A compatible PyTorch wheel must be selected for the target machine. Full audio-feature reconstruction may require system audio tooling such as `ffmpeg`, depending on the selected FunASR/torchaudio loading path.

5 Reproduction Procedure

The GitHub repository contains the final-only source code, scripts, tests, and runtime dependency specification. Install dependencies and download the public Hugging Face checkpoint to the expected project-relative model directory:

```
python -m pip install -r requirements.txt
cd artifacts/final_refit
hf download BertramM/whereisai-lingnan-track1-audio \
  --local-dir audio_delivery_pairwise_all_labeled_clean_no_quality_v1
cd ../../
```

5.1 Normal Inference

With organizer-provided Phase 1 assets and the corresponding final feature files available under `artifacts/`, generate predictions with:

```
PYTHON_BIN=python DEVICE=cpu bash scripts/142_infer_phase1_full_audio_delivery.sh
```

5.2 Full Feature Reconstruction

When starting from organizer-provided raw Phase 1 assets, build the final provided-text manifest and feature tables before normal inference:

```
PYTHON_BIN=python bash scripts/140_build_phase1_full_provided_text_manifest.sh
PYTHON_BIN=python bash scripts/141_build_phase1_full_provided_text_features.sh
PYTHON_BIN=python DEVICE=cpu bash scripts/142_infer_phase1_full_audio_delivery.sh
```

5.3 Exact Frozen-Submission Check

When the local frozen audit file is available, the following command also compares the output byte-for-byte with the frozen candidate:

```
PYTHON_BIN=python DEVICE=cpu bash scripts/155_reproduce_final_submission.sh
```

The available scripts are:

- `140_build_phase1_full_provided_text_manifest.sh`: build the provided-text Phase 1 manifest.
- `141_build_phase1_full_provided_text_features.sh`: extract final prosody, Emotion2Vec, and WavLM features and build the final feature table.
- `142_infer_phase1_full_audio_delivery.sh`: execute final inference with the released checkpoint.
- `155_reproduce_final_submission.sh`: run inference and compare the generated JSONL byte-for-byte with the frozen candidate when that local audit file is available.

`artifacts/final_refit/audio_delivery_pairwise_all_labeled_clean_no_quality_v1/`

6 Validation and Submission Coverage

The final code tests passed 15/15. Final inference passed both the submission format audit and the dependency audit. The frozen submission contains 530 rows:

Source	Questions	Answer distribution
GigaSpeech	200	A=119, B=81
MELD	210	A=86, B=62, C=62
EmoV-DB	120	A=64, B=56
Total	530	A=269, B=199, C=62

Table 3: Frozen Phase 1 submission coverage.

7 Submission Checklist

- The Hugging Face model repository is public and ungated.
- The released checkpoint and configuration hashes are recorded above.
- The public GitHub repository contains code, runtime specifications, tests, and this report.
- The competition form should contain the public Hugging Face URL and the captain name required by the organizers.